



**Verified Carbon
Standard**

SOLAR PV BESS AND CBG HYBRID PROJECT IN THAILAND



Document Prepared by Kosher Climate India Private Limited

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CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Sectoral Scope and Project Type	4
1.3	Project Eligibility	4
1.4	Project Design	5
1.5	Project Proponent	5
1.6	Other Entities Involved in the Project	5
1.7	Ownership.....	6
1.8	Project Start Date	6
1.9	Project Crediting Period	6
1.10	Project Scale and Estimated GHG Emission Reductions or Removals	6
1.11	Description of the Project Activity	7
1.12	Project Location	8
1.13	Conditions Prior to Project Initiation	8
1.14	Compliance with Laws, Statutes and Other Regulatory Frameworks	9
1.15	Participation under Other GHG Programs	9
1.16	Other Forms of Credit.....	9
1.17	Sustainable Development Contributions	10
1.18	Additional Information Relevant to the Project	10
2	SAFEGUARDS	11
2.1	No Net Harm	11
2.2	Local Stakeholder Consultation	11
2.3	Environmental Impact	13
2.4	Public Comments	13
2.5	AFOLU-Specific Safeguards	13
3	APPLICATION OF METHODOLOGY.....	13
3.1	Title and Reference of Methodology	13
3.2	Applicability of Methodology	14
3.3	Project Boundary	20
3.4	Baseline Scenario	21

3.5	Additionality	22
3.6	Methodology Deviations	30
4	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	30
4.1	Baseline Emissions	30
4.2	Project Emissions	36
4.3	Leakage.....	36
4.4	Net GHG Emission Reductions and Removals	36
5	MONITORING	37
5.1	Data and Parameters Available at Validation	37
5.2	Data and Parameters Monitored.....	39
5.3	Monitoring Plan.....	40

1 PROJECT DETAILS

1.1 Summary Description of the Project

The solar hybrid project involves installation of 49 MWp ground – mounted solar PV tracking system, 0.99 MW gas engine using Compressed Biogas (CBG) and 136.24MWh Battery Energy Storage System (BESS) located at Aranya Prathet District, Sa Kaeo Province, Thailand.

The purpose of this project activity is to generate electricity using renewable energy sources like by harnessing the solar energy by using of solar photovoltaic technology and using CBG for gas engine. In addition to the generation unit the project includes a BESS system to increase the efficiency. The energy generated from the project is purchased by electricity generating authority of Thailand (EGAT) under a Private Power Purchase Agreement (PPA) between Super solar energy company Ltd and EGAT.

The project is commissioned on 20-January-2023

Scenario existing prior to implementation of project activity:

The scenario prevailed prior to the implementation of the project activity, is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid connected power plants. The project activity helps to avoid CO₂ emission from fossil fuel-based power plants in the absence of the project activity and generate renewable electricity, which displaced part of the electricity otherwise supplied by grid connected power plants. The expected average CO₂ emission reduction from the project is 18,713 tCO_{2e}/annum

1.2 Sectoral Scope and Project Type

Switch of fossil fuel component:

Sectoral Scope : 01 - Energy industries (Renewable/non-renewable sources)

Project Type : Renewable energy projects

This is not AFOLU project and is not a grouped project.

1.3 Project Eligibility

The project activity reduces GHG emission levels included in six Kyoto Protocol greenhouse gases, which are listed under the scope of the VCS Standard version V4.5.

The project uses ACM0002. Ver. 21.0 – “Grid-connected electricity generation from renewable sources”. The methodology ACM0002. Ver. 21.0 is applicable to the project activities since it comprises renewable energy units, such as photovoltaic and CBG engine, that is supplying electricity to Electricity Generating Authority of Thailand (EGAT).

Since ACM0002 is approved methodology under an approved GHG program, the project is eligible under the scope of the VCS Program according to VCS Standard.

Furthermore, the project activity is not included in any of the excluded project activities, those listed under the Table 1 of VCS standard version V4.5.

Hence the project activity is eligible to register under the VCS program standard version 4.5.

1.4 Project Design

- ☒ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☐ The project is a grouped project

The project has been designed to include single location.

Eligibility Criteria

The project is not a grouped project. Thus, this section is not applicable.

1.5 Project Proponent

Organization name	Super Solar Hybrid Company Limited
Contact person	Sunee MOUNGCHAREON
Title	-
Address	223/61, 14 th floor, Country Complex Tower A, SunpawutRoad, Bangnatai, Bangna, Bangkok 10260 Thailand.
Telephone	+662-361-5599
Email	Sunee.m@supercorp.co.th

1.6 Other Entities Involved in the Project

Organization name	Kosher Climate India Private Limited
Role in the project	Project Representative
Contact person	Vamsi Krishna
Title	Global Business Head
Address	Zee Plaza, No.1678, 27 th Main Road, Sector 2, HSR Layout, Bengaluru, Karnataka 560102, India
Telephone	080-25720814
Email	vamsi@kosherclimate.com

1.7 Ownership

Super Solar Hybrid Company Limited is the project owner of SPP Hybrid Project. They have the legal right to control and operate the project activities. The engineering, procurement and construction (EPC) agreement, commissioning certificate (CC), Operation and maintenance certificate are evidences for the ownership of the plant equipment and power generating.

1.8 Project Start Date

The project start date is 20-january-2023, as per the commissioning certificate (CC) the company has started supplying electricity to Electricity Generating Authority of Thailand (EGAT) from this date.

1.9 Project Crediting Period

The VCS Standard 4.5 states that the project crediting period for non-AFOLU projects shall be either seven years, twice renewable for a total of 21 years, or 10 years fixed. Since this project is a non-AFOLU project, the renewable crediting period of 7 years, twice renewable for a total of 21 years is chosen by the Project Proponent (PP).

The first crediting period starts on 20-january-2023 and ends on 19-january-2030.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☒ <20,000 tCO₂e/year
- ☐ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year

☐ >1,000,000 tCO₂e/year

Project Scale	
Project	
Large project	X

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	19,020
Year 2	18,917
Year 3	18,814
Year 4	18,713
Year 5	18,611
Year 6	18,510
Year 7	18,410
Total	130,995
Total number of crediting years	7
Annual Average over the crediting period	18,713

1.11 Description of the Project Activity

The project activity involves hybrid energy generation from PV plant and CBG engine along with a BESS system. The proposed PV plant will incorporate bifacial mono-crystalline silicon technology solar modules mounted in a single axis (N-S) tracking system a total installed DC capacity of 48.9 MWp.

The selected module is the Risen Solar RSM132-8-640BMDG, rated at 640Wp and the plant will integrate 193 sungrow SG250HX string inverters with 12 MPPTs with maximum efficiency 99%

The battery used in the project activity is based on Li-ion technology of Sungrow. Gas engine generator with capacity of 0.99 MW is used to supplement the electricity supply from solar PV system and BESS. The gas engine will use compressed bio-methane gas (CBG) as fuel. The CBG will be supplied from Northern Biogas Company Limited (NBL) factory located in Nakhon Ratchasima province.

The main technical details of SPP Hybrid project are shown in the table below:

Technology	Type	Manufacturer	Model
Solar PV System			
PV Module	Mono PERC (640 W)	Risen Energy	RSM132-8-640BMDG
Inverter	String inverter	Sungrow	SG250HX
Transformer	6,750kVA, 0.8/33kVA Transformer	Sungrow	Oil immersed, self-cooled hermetically sealed.
	25MVA, 33/115kV Transformer	Daihen Electric	
Mounting Structure	Single axis tracking system	Archtech Solar	Skysmart II 2V50-1 / 2V33
BESS System			
Battery, PCS, EMS & SCADA System	Lithium iron phosphate (LFP)	Sungrow	45MW / 136.24MWh
Compress Biogas System			
Gas Engine	Type 3 JMC 320 GS engine	GE Jenbacher	JMC 320 GS-L. L

1.12 Project Location

The project is located in Aranya Prathet District, Sa Kaeo Province, Thailand which is in the eastern part of Thailand.

Project Location	Latitude	Longitude
Aranya Prathet District, Sa Kaeo Province, Thailand	N 13°47'11.60"	E 102°26'02.45.

1.13 Conditions Prior to Project Initiation

The project is Greenfield project. There was no project installed prior to commissioning of this project. The scenario existing prior to the implementation of the project activity, is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid

connected power plants (Which is predominantly based on fossil fuels) and by the addition of new generation sources.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Energy Industry Act, B.E. 2550 (2007)¹ in Thailand also known as the Energy Industry Act of 2007, regulates the power sector in Thailand. Even though the act encourages the development of renewable energy sources such as solar, wind and biomass, it does not mandate electricity generation from solar projects. Hence, no national or sectoral policies prevent the implementation of the project activity.

The project as in the project activity are not mandated by law or regulations and are entirely a voluntary action.

The project activity is in compliance with all the relevant statutory and regulatory laws applicable in the host country.

Relevant laws, statutes, and regulatory frameworks summarized as below:

- Energy Regulatory Commission Announcement
- Ministry of Industry Announcement
- Factory Act, B.E. 2535 (1992)²

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

Not Applicable. The project activity has applied for registration under VCS programme only and hence GHG emission reductions and removals generated by the project will not be used for compliance under such programs or mechanisms.

1.15.2 Projects Rejected by Other GHG Programs

No, the project activity is not rejected by other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

¹ https://climate-laws.org/documents/energy-industry-act-b-e-2550_bf19?id=energy-industry-act-b-e-2550_154f

² <https://faolex.fao.org/docs/pdf/tha209657.pdf>

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes

☒ No

Supply Chain (Scope 3) Emissions

The project is generation of electricity from solar energy and supply of electricity to the grid and the project does not impacts any emissions associated with a good or service of any specific organisation. Hence, this section is not applicable.

1.17 Sustainable Development Contributions

The project activities directly cover key areas of sustainable development including renewable power generation, innovative technological integrations, efficient usage, environmental well-being, economic and social well-being by generating permanent and temporary employment opportunity within the vicinity of the project.

Among 17 SDGs,

1. Affordable and clean energy (SDG-7) by 16MWac renewable power generation.
2. Decent work and economic growth (SDG – 8): Almost 250 employees were employed during the construction phase of the project and almost 25 employees are working during operation phase.
3. Climate action (SDG – 13): Reduction of 18,713 tons of CO₂ eq. greenhouse gas (GHG) emission per year by coal replacement.

1.18 Additional Information Relevant to the Project

Leakage Management

Not Applicable

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not Applicable

2 SAFEGUARDS

2.1 No Net Harm

The project activity doesn't provide any potential negative environmental and socio – economic impacts. As the project involves renewable energy technology like solar for electricity generation, it results in zero GHG emissions. Solar power project activity operations do not result in direct air pollution, noise pollution. Thus, there is no any significant impact due to implementation of project activity on air, water, soil quality and ambience are envisaged due to the project activity.

2.2 Local Stakeholder Consultation

The details of stakeholder meetings are as follow:

Date of meeting: 29- January- 2021

Location: The office Aranyaprathet district Sa Kaeo Province.

The project officials visited the project area from 2-december- 2020 to 4-december-2020 to meet the community leaders in the study area with the aim to convey the details of the project to the public and relevant agencies are informed, about the format for organizing the public participation activities to carry out project. Conducted the meeting to listen to public opinions on the project implementation and measures to prevent, correct and monitor the project's environmental impacts on 29-january-2021 from 9 AM to 12PM at the meeting room of Aranyaprathet District Office. Sa Kaeo Province. Purpose of the meeting, details of the project, draft measures to prevent, correct and monitor environmental impacts of the project were presented in the meeting. Meeting concluded with presentation of summaries, comments and suggestions from the attendees. Total 158 people attended the meeting. The project officials have compiled and summarized the recommendation from meeting participants including clarification of information on each issue and improvement of preventive measures to be appropriate and consistent with community concerns. There was an opening to listen to additional opinions from the public after organizing meetings for a period of at least 15 days, carried out from 30-January-2021 to 15-Febrary-2021 through communication channels, including telephone, fax, email, and postcards at attached together with documents published on the day of the meeting, proceed with posting report summarizing the results of the meeting to listen to opinions, etc on the public relations board of the community and various agencies such as local

government agencies, community offices, community hall, etc for a period of 15 days from 18-Febrary-2021 to 4- march-2021.

Suggestions and concerns from stakeholder and the clarifications given are summarised in the table below:

Suggestions/Concerns	Opinion giver	Clarification and related measures
Establishing a solar power plant will there be an increase in heat in the area? Does it affect the season or not?	General public, village no.1, Ban Han Sai Subdistrict Administrative Organization, Aranyaprathet District, Sa Kaeo Province.	The project's electricity production is in the form of solar energy. t will use solar energy panels also known as "solar cells, they absorb heat energy from the sun or light intensity that is diffused down to the ground in every natural daylight hours to change the form of heat energy is converted into electrical energy. Therefore, the use of solar panels absorbing heat energy from the sun is therefore not a factor that effects the season. But on the other hand, it can be considered that electricity production in the use of solar energy is clean energy and helps reduce global warming when compared to combustion- based electricity generation.
Project implementation may affect lighting and reflects stability due to the project	Head of the industrial factory group industrial office Sa Kaeo Province.	Project implementation requires the use of solar energy panels where there is glass or film installed, it must

area; it may be the route of a government aircraft.		prevent effective reflection of light. To prevent reflection of light up to 95% to make the cell panels absorb as much solar energy as possible.
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All the questions raised by the stakeholders were clarified by the project owner. No concerns raised by the stakeholders that require change in project design

2.3 Environmental Impact

The project activity has no significant impact on the environment.

2.4 Public Comments

This section will be updated after the commenting period is completed.

2.5 AFOLU-Specific Safeguards

Not Applicable as the project is not a AFOLU project

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The project activity is a large-scale project with capacity of power generation more than 15 MW as per CDM terminology. Accordingly, the methodology applied for the is approved methodology for large - scale CDM project activities. The details of the methodology are as follows:

Methodology	: ACM0002
Project Type	: Type-1: Renewable Energy Projects
Title	: Grid-connected electricity generation from renewable sources ³
Version	: Version 21.0; EB 116

³ <https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL01OU2RYC6N3HASIXV7K84QBG9>

Reference : CDM Methodologies⁴

Tools involved in the project are listed below:

1. Tool 07: “Tool to calculate the emission factor for an electricity system” - Version 07.0⁵
2. Tool 01: “Tool for the demonstration and assessment of additionality” – Version 07.0⁶
3. Tool 27: “Tool for the Investment analysis” Version - 12.0⁷
4. Tool 24: “Common practice” Version – 03.1⁸

3.2 Applicability of Methodology

Methodology applicability criteria of the project

Methodology Applicability Criteria	Project applicability
This methodology is applicable to grid – connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	The proposed project activity is a green field, that is supplying electricity to Electricity Generating Authority of Thailand (EGAT). Therefore, it confirms to the said criteria
In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant;	The project is a solar hybrid power project consists of 3 major systems the ground-mounted solar PV tracking system, the biogas system and the battery energy storage system (BESS).

⁴ <https://cdm.unfccc.int/methodologies/DB/XKCRT4QQUUWXXZMQRXUGESOWON451M>

⁵ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

⁶ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

⁷ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

⁸ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf>

(b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	The project is integrating a 45 MW BESS. Therefore, it confirms to the criteria of integrate BESS with a Greenfield power plant.
The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit;	The project activity is the installation of a new grid connected renewable solar PV tracking system and biogas system. Thus, it meets the first applicability condition.
(b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity;	The proposed project activity is the installation of a new solar hybrid project. Therefore, the said criteria is not applicable.
(c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents);	As per the final technical due diligence report, BESS is an integral part of the design of the project activity. Therefore, the said criteria is applicable.
(d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be	Under normal condition BESS is charged with electricity generated from the project only.

accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	If BESS is charged using grid or fossil fuel electricity generator during crediting period, then project emission will be calculated as per TOOL05 or TOOL07 respectively. Thus, the said criteria is applicable.
In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	The project activity is not hydro power plant. Thus, the said criteria is not applicable.

In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	The project activity is not hydro power plant. Thus, the said criteria is not applicable.
The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The proposed project activity is a greenfield hybrid solar project involving ground mounted solar PV, BESS and CBG generator for electricity generation. It doesn't involve switching from fossil fuel to renewable energy source or biomass fired power plants/units. Thus, the said criteria is applicable.
In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	The project activity is a greenfield project. Thus, the said criteria is not applicable.

Tool 07: Tool to calculate the emission factor for an electricity system - Version 7.0, applicability of the project

Applicability criteria	Project applicability
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to the grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	The project activity is a greenfield solar power generation plant and hence, according to the applied methodology, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system”. Thus, the said criteria are applicable.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, the conditions specified in “Appendix 2: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Since the project activity is grid connected, this condition is applicable and the emission factor has been calculated accordingly. Thus, the said criteria are applicable.

In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country	The project activity is located in Thailand, a non-Annex I country. Therefore, this condition is not applicable for the project activity.
Under this tool, the value applied to the CO ₂ emission factor of bio fuels is zero	The project activity is a grid connected solar power project and doesn't involve utilization of bio fuels, so the condition is not applicable for the project activity.

Tool 01: Tool for the demonstration and assessment of additionality – Version 07.0, applicability of the project.

Applicability criteria	Project applicability
The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool	The project proponent doesn't propose any new methodologies for the project and thus doesn't propose alternative methods to demonstrate additionality for consideration by the executive board. The project proponent is following methodological tool 01 version 07.0 to prove additionality.
Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	The project activity is a large-scale solar hybrid project The project proponent is using methodological tool 01 “Tool for the demonstration and assessment of additionality” – version 07 as per the methodology ACM0002 – version 21 to prove the additionality.

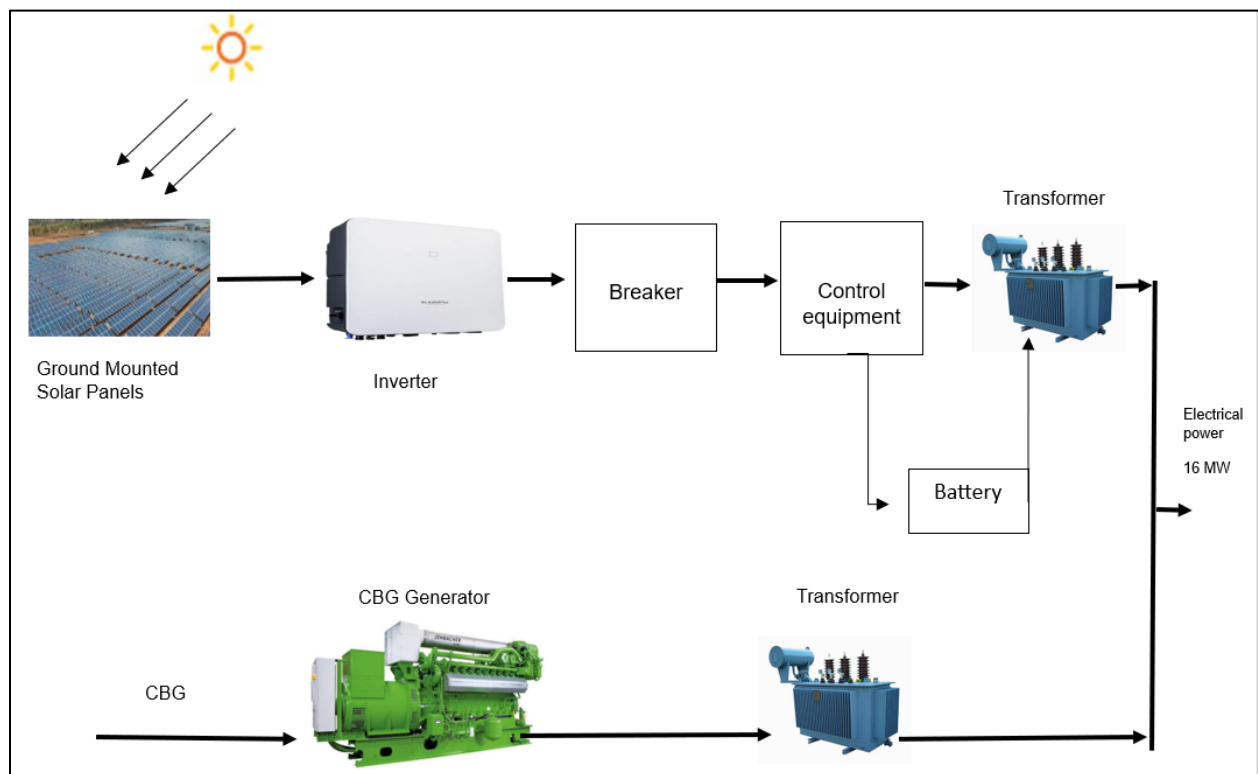
TOOL27: Investment analysis version 12.0 scope and applicability of the project

Applicability criteria	Project applicability
This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines	The project proponent is using methodological tool 01 “Tool for the demonstration and assessment of additionality” – version 07 as per the methodology ACM0002 – version 21 to prove the additionality. Thus the criterion is applicable.

“Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	
In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied methodology ACM0002 does not contain requirement for the investment analysis. Hence this criterion is not applicable

3.3 Project Boundary

As per ACM0002 version 21 para 22 “The spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the CDM project power plant is connected to” Thus, the project boundary includes the ground mounted solar PV modules, BESS, CBG generator, other equipments and transmission systems.



Project Boundary

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from electricity generation	CO ₂	Yes	Major Source. Electricity generation emission is included in electricity grid-based emissions.
		CH ₄	No	Excluded for simplification, minor emission source
		N ₂ O	No	Excluded for simplification, minor emission source.
		Other	---	
Project Activity	Greenfield Solar Power Project activity	CO ₂	NO	No CO ₂ emissions are emitted from the project
		CH ₄	NO	Project activity does not emit CH ₄
		N ₂ O	NO	Project activity does not emit N ₂ O
		Other	---	

3.4 Baseline Scenario

The baseline scenario of the project activity is established using the methodology ACM0002 “Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources”- Version 21.0 and with Tool 07, “Tool to calculate the emission factor for an electricity system “- Version 7.0.

According to methodology para 24 “If the project activity is the installation of a Greenfield power plant with or without a BESS, the baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07”

The project activity involves setting up a hybrid facility which includes 49 MWp ground-mounted solar PV tracking system, 0.99 MW biogas generator and 45 MW battery energy storage system (BESS) components and electricity generating authority of Thailand is purchasing the electricity generated through a contracted power purchase agreement for 20 years.

As per methodology ACM0002 version 21.0, para 47 “Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plant”

$$BE_y = EG_{PJ, y} \times EF_{grid, CM, y}$$

Where:

BE_y = Baseline emissions in year y (t CO₂/yr)

$EG_{PJ, y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid, CM, y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO₂/MWh)

3.5 Additionality

The additionality of a VCS Project shall be demonstrated by applying the following approach having two ~~components~~

- (i) A Legal Requirement Test; and
- (ii) An Additionality Test.

(i) Legal requirement test

The Energy Industry Act, B.E. 2550 (2007)⁹ in Thailand also known as the Energy Industry Act of 2007, regulates the power sector in Thailand. Even though the act encourages the development of renewable energy sources such as solar, wind and biomass, it does not mandate electricity generation from solar projects. Hence, no national or sectoral policies prevent the implementation of the project activity. The project as in the project activity are not mandated by law or regulations and are entirely a voluntary action. The project activity is in compliance with all the relevant statutory and regulatory laws applicable in the host country.

⁹ https://climate-laws.org/documents/energy-industry-act-b-e-2550_bf19?id=energy-industry-act-b-e-2550_154f

The project activity does not fulfil the criteria of positive list as provided in CDM Tool 32: “Methodological Tool – Positive List of Technologies” and hence additionality of the project activity is demonstrated through a project specific additionality test.

(ii) Additionality Test

Additionality has been demonstrated as per the applied methodology ACM0002 (Version 21.0). Methodology requires the project owner to determine the additionality based on “Tool 01: Tool for the demonstration and assessment of additionality”, Version 7.0.0.

The step-wise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:

As per the applied methodology requirement, Additionality of the project activity is demonstrated using the Methodological tool “Tool 01: Tool for the demonstration and assessment of additionality” Version 07.0.0. The tool defines the following steps:

Sub Step 0: Demonstration whether the project activity is the first-of-its-kind.

The project activity is not the first of its kind as implementation of solar power project in the province is not first of its kind

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations.

Sub-step 1a: Define alternatives to the project activity

As per the applied ACM0002 version 21.0; Para 24, if the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid connected power plant and by the addition of new generation sources.

However, for the assessment of additionality the following alternatives are identified:

Alternative 1: The project activity undertaken without being registered as a VCS project activity.

The project owner could proceed with the implementation of the project without considering the carbon credit benefits. The electricity produced from the renewable energy project would have been sold to grid. The is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, the project activity is not feasible without revenues from the sale of carbon credits. This argument has been discussed in step 2 of the Additionality section.

Alternative 2: No project activity is undertaken and equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid i.e., Continuation of the present situation.

The project owner would have continued without investment in the project activity with usual business activity. The electricity delivered to the grid by the project activity would be generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid. It means that a power plant with emission factor equivalent to grid mix would have been

supplied electricity in absence of new project plant or added capacity. Grid emission factor is a reasonable benchmark that provides the proxy performance of the baseline power plant.

Outcome of Sub-step 1a:

All the realistic alternative for the project activity have been enlisted above.

Thus, the two alternatives are mentioned above as per step of additionality tool, the first alternative is not possible as project activity is not viable without the carbon credit benefits and second alternative is the baseline scenario for the project activity as per methodology as mentioned in the section 3.4 of the PD.

It is to be noted that being the Greenfield project activity, “the baseline scenario is electricity delivered to the grid by the project activity would have been generated by the operation of the grid-connected power plant and by the addition of new generation sources, as reflected in the combined margin (CM) calculation describe in the “Tool 07: Tool to calculated emission factor for an electricity system”.

Sub-step 1b: Consistency with mandatory laws and regulations:

The alternative(s) shall be in compliance with all applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions. e.g., to mitigate local air pollution.

Following national and sectoral policies and regulations are applicable for the project activity. Project owner has considered all the applicable national and sectoral level policies in demonstrating the regulatory compliance of the project and baseline scenario.

National/sectoral policies and regulations:

- The Energy Industry Act, B.E. 2550 (2007)¹⁰
- Energy Regulatory Commission Announcement
- Ministry of Industry Announcement
- Factory Act, B.E. 2535 (1992)¹¹

According to all the referred policies and regulations the baseline scenario is in compliance with all the applicable legal and regulatory requirements.

Outcome of sub-step 1b

As per the above analysis both the alternatives enlisted above are found to be comply with the mandatory laws and regulation of the host country. Since, there is no law mandated the project owner to invest and implemented the project activity. The project owner has a choice to go for either:

Alternative 1: The project activity undertaken without being registered as a VCS project activity.

Or

¹⁰ https://climate-laws.org/documents/energy-industry-act-b-e-2550_bf19?id=energy-industry-act-b-e-2550_154f

¹¹ <https://faolex.fao.org/docs/pdf/tha209657.pdf>

Alternative 2: No project activity is undertaken and equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid i.e., Continuation of the present situation.

Step 2: Investment Analysis

As per para 29 of “Tool 01: Tool for the demonstration and assessment of additionality” it is determined that the project activity is not an economically attractive or financially feasible option.

To conduct the investment analysis, Methodological tool 27: Investment analysis, version 12.0, EB 116 Annex 02 has been referred

Sub-step 2a: Determine appropriate analysis method

As per “Tool 01: Tool for the demonstration and assessment of additionality” (version 07.0.0), for financial analysis of the project, the following three options are available:

Option I: Simple Cost Analysis

Option II: Investment Comparison Analysis

Option III: Benchmark Analysis

The project activity envisages to import and export the power to electricity generating authority of Thailand (EGAT) and the revenues from the sale of electricity would be generated in accordance with the terms and tariffs established in Power Purchase Agreement (PPA). Thus, simple cost analysis (Option I) cannot be used as the analysis method as the sale of the units of generation electricity shall result in a revenue stream during the operations of the project activity.

In the absence of the project activity grid electricity would have been the obvious choice for the project, which requires no investment. Hence, investment comparison analysis (option II) is also not appropriate for the project activity.

However, after eliminating Option I and Option II, the use of the Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project revenue during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be employed for analysis of the said project.

Sub-step 2b: Option III. Apply benchmark analysis.

According to the “Tool 01: Tool for demonstration and assessment of Additionality”, the financial indicator can be based either on (1) Project IRR or (2) Equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for the project activity has been as post tax equity IRR.

As per Tool 27: Investment analysis, version 12.0, it required/expected returns on equity are appropriate benchmarks for a post-tax Equity IRR. The post-tax Equity IRR is considered as the financial indicator and the benchmarks used is cost of equity. Hence, the benchmarks used are applicable to the project activity and the type of IRR calculation presented.

Selection of Appropriate Benchmark

The benchmark has been considered in accordance with Guidance 19 of EB 112 Annex 2, “The values in the table in the Appendix may also be used, as a simple default option”.

Methodology deployed for arriving at a suitable value of Benchmark using Default Value has been described below:

- As the proposed project activity generates power utilizing solar energy, Group 1 as per para 5a of Appendix of EB 116 Annex 2 has been identified as a suitable category.
- The investment analysis has been carried out in Nominal terms. Accordingly, Default value as given in table under the Appendix, Tool 27 has been adjusted by adding suitable forecasted inflation rate taken from IMF world Economic database as no inflation forecast or inflation target published by Bank of Thailand (Central Bank of Thailand).
- Average forecasted inflation rate for the host country (Thailand) published by the IMF (International Monetary Fund) World Economic Outlook for the next five years has been considered

The decision-taken for the project is on 18-January-2021. Hence applicable inflation rate has been chosen from the IMF databases accordingly for the estimation of resultant benchmark.

The benchmark has been computed in the following manner:

Default Value Benchmark:

The Cost of Equity has been considered using the “Methodological tool: Investment analysis”.

Default value in the Tool 27: Investment analysis, version 12.0:

Table under Appendix in EB116, Annex 2 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Thailand = 8.93%¹²

Thus, minimum cost of equity considered for calculation of Benchmark = 8.93%

The Required return on equity (benchmark) was computed in the following manner:

$$\text{Nominal Benchmark}^{13} = \{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$$

Where:

Default value for Real Benchmark = 8.93% (as per Appendix of EB 116, Annex 2)

¹² <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

¹³ As per Pg. 320 of Corporate Finance, Second Edition of Aswath Damodaran

Inflation Rate forecast for by (International Monetary Fund) World Economic Outlook database for Thailand.

5-year inflation Forecast for Thailand as IMF World Economic Outlook available at the time of investment decision and corresponding benchmark values is

Inflation Rate for 5 years	Source
2.60%	World Economic Outlook database: April 2022

Corresponding benchmark values applicable at the time of investment decision time: Nominal Benchmark = $\{(1+8.93\%) * (1+2.60\%)\} - 1 = 11.76\%$

Inflation Forecast	Benchmark
5 years	5 years
2.60%	11.76%

Sub-step 2c: Calculation and comparison of financial indicators

The period considered for Post Tax Equity IRR calculations is 25 years, which corresponds to the operational lifetime of the project activity.

Depreciation, and other non-cash items related to the project activity, which have been deducted in estimating gross profits on which tax is calculated, is added back to net profits for the purpose of calculating the financial indicator.

Varying and fixed Input values considered for the IRR calculation are provided below

INPUT PARAMETERS			
Particulars	Value	Unit	Source/Remarks
Capacity of the project	16.00	Mw	PIM
Annual Net generation from BESS	81.0880	GWh	Calculated
Degradation	0.60%	-	PIM
Project cost	2072.93	Million TBH	PIM
Debt	70%	%	PIM
Equity	30%	%	PIM
Debt	1451.05	Million TBH	Calculated
Equity	621.88	Million TBH	Calculated
Interest rate	7.00%	%	PIM
Debt Repayment tenure	13	years	PIM
Moratorium	0	year	PIM
Operation and Maintenance (% of project cost)	0.97%	%	PIM
Operation and Maintenance	20.1	Million TBH	PIM

Escalation in O & M	4%	%	PIM
CBG Cost per kg per day	4.25	TBH/kg/day	PIM
Amount of CBG purchased	30000	kg/day	PIM
CBG Cost per year	46.5375	Million TBH/year	Calculated
Fixed Tariff	1.81	TBH/kWh	PIM
Fixed Tariff (after discount of 43.12%)	1.029528	TBH/kWh	Calculated
Variable Tariff (2021)	1.85	TBH/kWh	
Inflation on Variable Tariff	2.60%	%	World Economic Outlook database: April 2022
Tariff applicable	2.928	TBH/kWh	Calculated
Depreciation Rate	4.00%	%	PIM
Corporate Tax 0-8 years	0.00%	%	PIM
Corporate Tax 8 th year onwards	20%	%	PIM

Post Tax Equity IRR for the project against the benchmark value is shown in table below. Thus, it is evident that the project is not financially attractive as the equity IRR is less below the benchmark value.

Project IRR	Benchmark
7.96%	11.76%

Sensitivity Analysis

The robustness of the conclusion drawn above, namely that the project is not financially attractive, has been tested by subjecting critical assumptions to reasonable variation. As required by Annex 2 of EB 116, only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation. PP has identified the total revenue from the project activity is dependent on the Tariff, Plant generation, Project Cost and O&M Costs constitute more than 20% of the project costs. These factors have been subjected to a 10% variation on either side and the results of the sensitivity analysis indicate that even after applying such variation the IRR does not cross the benchmark.

Variation %	-10%	Normal	10%	Variation required to reach benchmark	Value required to reach benchmark
Tariff	5.88%	7.96%	10.00%	18.70%	3.475
Generation	5.78%	7.96%	10.14%	17.40%	95.197312
Project Cost	9.96%	7.96%	6.30%	-17.50%	1710.167567
O&M Cost	8.16%	7.96%	7.76%	NA	NA

An analysis has been done to identify the percentage variation at which the financial indicators will equal/breach the benchmark and the probability of its occurrence. Based on sensitivity analysis it can be concluded that the proposed project activity is additional with reasonable variation in values and is not likely to reach the benchmark value. The occurrence of these events is unlikely for the following reasons:

a) **Tariff:** The Tariff rate of electricity used for investment analysis i.e., tariff 2.928 THB/kWh is sourced from the project information memorandum (PIM) applicable at the time of investment decision. The project will breach the benchmark value at a tariff variation of 18.70% and the value required to reach benchmark is 3.475 THB/kWh. However, the actual tariff based on the PPAs signed is same as the tariff considered in the IRR calculation i.e., 2.928 THB/kWh ~~1644 VND/kWh~~ fixed for PPA tenure without any escalation which is much below the tariff value required to breach the benchmark value. Hence, increase in tariff is unlikely.

b) **Generation:** The generation used for investment analysis i.e., 81.088 GWh is sourced from the project information memorandum (PIM) applicable at the time of investment decision and the IRR breach the benchmark value at a generation variation of more than 17.40% increase in the estimated generation, the value of generation required to reach benchmark is 95.197312 GWh. The increase in generation value to breach the benchmark is highly unlikely as the estimated generation is based on the annual solar energy radiation estimated for project lifetime and given the analysis above it's highly unlikely that generation will increase above breaching value.

c) **Project Cost:** The project cost considered for investment analysis i.e., 2072.93 Mn THB. The cost is sourced project information memorandum (PIM) applicable at the time of investment decision. A variation of -17.50% and a value of 1710.167567 Mn THB are required for IRR to breach benchmark which is not possible as the project is already commissioned at the estimated cost in the IRR which can be verified from the EPC contracts. The actual cost incurred in commissioning of the project is close to the estimated cost. Hence reduction of project cost is unlikely to happen.

d) **O&M Costs:** The sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is subject to escalation (as evidence by the O&M agreement) and also subject to inflationary pressure, any reduction in the O&M costs is highly unlikely.

Step 3: Barrier analysis

Barrier analysis has not been used

Step 4: Common practice analysis

In the geographical boundary of the project (Thailand) before the start date of the project no other hybrid projects which includes ground mount PV, BESS and CBG generator were commissioned.

3.6 Methodology Deviations

There is no methodology deviation

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

As per methodology ACM0002, version 21, para 47 “Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants”

The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where

BE_y = Baseline emissions in year y (t CO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO₂/MWh)

As per para 49 of the methodology ACM0002 version 21 “If the project activity is the installation of a Greenfield power plant with or without the BESS, then:

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

As per the methodology combined margin grid emission factor has been calculated as per the “**Tool to calculate the grid emission factor for an Electricity System**” version 7.0

Thailand Grid Emission Factor for GHG Reduction Project/Activity dated 28-September-2017 Published by Thailand Greenhouse Gas Management Organization (Public Organization) has been used for the calculation of emission factor.

As per the "Tool to calculate the emission factor for an electricity system" Version 7.0, the following steps have been followed.

- (a) **Step 1:** Identify the relevant electricity systems;
- (b) **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional);
- (c) **Step 3:** Select a method to determine the operating margin (OM);
- (d) **Step 4:** Calculate the operating margin emission factor according to the selected method;
- (e) **Step 5:** Calculate the build margin (BM) emission factor;
- (f) **Step 6:** Calculate the combined margin (CM) emission factor.

Step 1: Identify the relevant electricity systems

For the purpose of determining the electricity emission factor, the project electricity system is defined as the electricity transmission system of Thailand which is a single system connected by transmission lines throughout the country and owned by the Electricity Generating Authority of Thailand (EGAT). Electricity imports from a connected electricity system are included and as per “Tool to calculate emission factor of an electricity system, for the purpose of determining the operating margin emission factor”, Version 07.0 is applied.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

Only grid connected power plants are included in the calculation, as per Option I of the “Tool to calculate the emission factor for an electricity system” version 07.0

Step 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods, which are described under Step 4:

- (a) Simple OM; or
- (b) Simple adjusted OM; or
- (c) Dispatch data analysis OM; or

(d) Average OM.

The simple OM method (Option a) is used for this study as the low-cost/must-run resources (LC/MR) constitute less than 50% of total electricity production. The share of low-cost/must run power plants in the last two years are in the range of 4.55% to 6.69% as shown in the below table.

Year	Production of total electrical energy (GWh)	LC/MR Analysis			
		Hydro	RE	Total LC/MR	% of LC/MR
2012	166,446	8,431	2,701	11,132	6.69
2013	164,826	5,412	3,427	8,839	5.36
2014	168,685	5,164	3,993	9,157	5.43
2015	169,040	3,724	4,230	7,954	4.71
2016	169,168	3,019	4,685	7,704	4.55

Data Source: EGAT (2017) ¹⁴

As per tool to calculate emission factor for an electricity system (Version 07.0), The simple OM method (option a) can only be used if low-cost/must-run resources constitute less than 50% of total grid generation in: 1) average of the five most recent years, or 2) based on long-term averages for hydroelectricity production. Since the low cost/must run resources constitute less than 50% of total grid generation as seen from the average of five most recent years, the Simple OM method can be used to calculate the Operating Margin Emission factor.

PP has chosen ex ante option, thus, no monitoring and recalculation of the emissions factor during the crediting period is required. PP has considered a data vintage of 3-year generation-weighted average, based on the most recent data available at the time of submission of the CDM-PDD to the Verifier for validation

Step 4: Calculate the operating margin emission factor ($EF_{grid, OMSimple,y}$) according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units. The simple OM may be calculated:

¹⁴ https://ghgreduction.tgo.or.th/images/Grid_Emission_Factor_2559_-_Finalised.pdf

Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit; Or
Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Thailand Greenhouse Gas Management Organization (Public Organization) has published the Thailand Grid Emission Factor for GHG Reduction Project/Activity on 28-September-2017 based on detailed authenticated information obtained from EGAT. This provides information about the Combined Margin Emission Factor of Thailand national grid. The Combined Margin is calculated ex ante using the guidelines provided by the UNFCCC in the “Tool to calculate the emission factor for an electricity system, Version 07.0”. We have, therefore, used the Combined Margin data published in the Grid Emission Factor for GHG Reduction Project/Activity, for calculating the Baseline Emission Factor.

As per “Tool to calculate the emission factor for an electricity system”, Option B (“Calculation based on total fuel consumption and electricity generation of the system”) is used to calculate simple OM emission factor. Where Option B is used, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{\text{grid,OM simple,y}} = \frac{\sum_i FC_{i,y} \times NCV_{i,y} \times EF_{\text{CO}_2,i,y}}{EG_y}$$

Where:

EF _{grid,OM simple,y}	Simple operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
EG _{m,y}	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
FC _{i,y}	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)
NCV _{i,y}	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
EF _{CO₂, i,y}	CO ₂ emission factor of fuel type i in year y (t CO ₂ /GJ)
EG _y	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh)

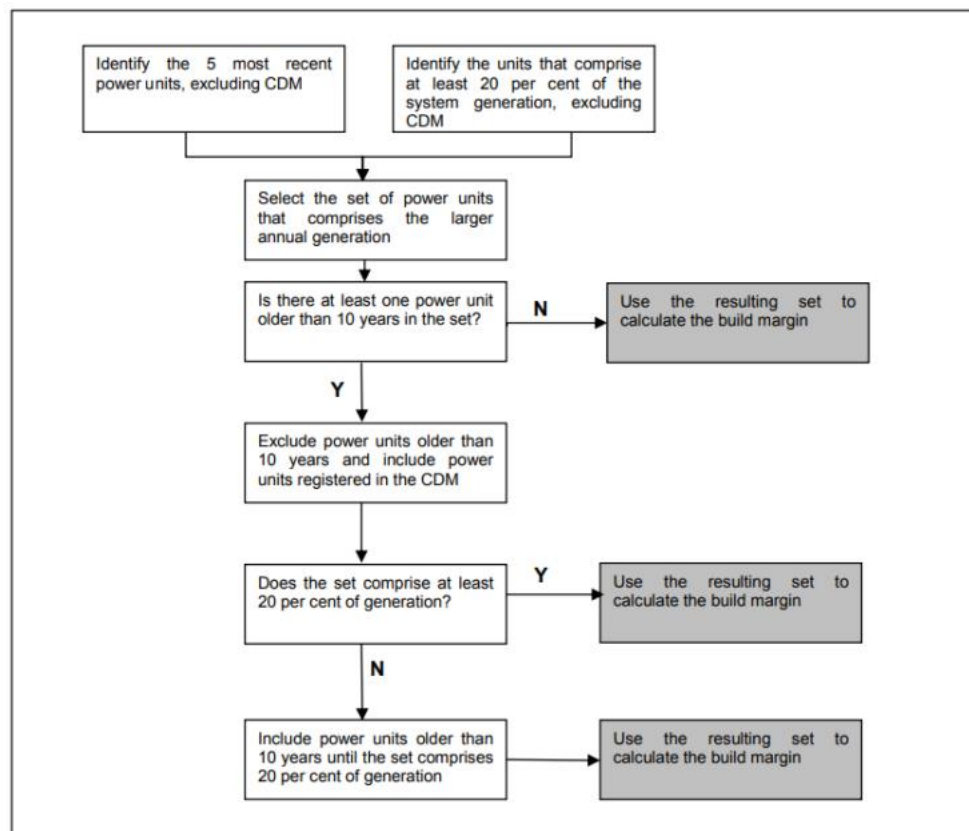
i	All fuel types combusted in power sources in the project electricity system in year y
y	the relevant year as per the data vintage chosen in STEP 3

As per the Thailand emission factor database published on 28-September-2017 weighted average operating margin is as below:

$$EF_{\text{grid,OM},y} = 0.5719 \text{ tCO}_2/\text{MWh}$$

STEP 5: Calculate the build margin emission factor ($EF_{\text{grid,BM},y}$)

The project participants have chosen Option I, i.e., fixing build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM PDD submission to the Verifier for validation. The build margin emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of a sample group of power units, during the most recent year y for which power generation data is available. The Sample group of power units m used to calculate the build margin should be determined via the procedure summarized in the diagram of the Tool.



Following this procedure, AEGSET >/20% is larger than AEGSET 5-units and all of these power units started supplying electricity to the grid less than 10 years ago, therefore AEGSET >/20% is applied as power units m for the Build Margin Using the equation given in the step 5 for the OM calculation, the Built margin is calculated for the year 2016 is as below:

$$EF_{\text{grid,BM},y} = 0.5609 \text{ tCO}_2/\text{MWh}$$

STEP 6: Calculate the combined margin (CM) emissions factor

Combined Margin – The combined margin is the weighted average of the simple operating Margin and the build margin. In particular, for intermittent and non-dispatch able generation types such as wind and solar photovoltaic, the Tool to calculate the emission factor for an electricity system, Version 7.0.0, EB 100, Annex 4, allows to weigh the operating margin and Build margin at 75% and 25%, respectively for wind and solar projects and 50% and 50%, respectively for hydro and biomass projects.

The baseline emission factor is calculated using the combined margin approach as described in the following steps:

Calculation of Baseline Emission Factor EF_y

The baseline emission factor EF_y is calculated as the weighted average of the Operating Margin emission factor ($EF_{\text{grid,OM},y}$) and the Build Margin emission factor ($EF_{\text{grid,BM},y}$):

$$EF_y = w_{\text{OM}} * EF_{\text{OM},y} + w_{\text{BM}} * EF_{\text{BM},y}$$

Where,

w_{OM} 75% weight for solar energy projects

w_{BM} 25% weight for solar energy projects

$EF_{\text{OM},y}$ calculated as described in Steps 3&4 above (tCO₂/MWh)

$EF_{\text{BM},y}$ calculated as described in Steps 5 above (tCO₂/MWh)

Baseline Emission factor (THAILAND National Grid) = 0.75*0.5719 + 0.25*0.5609

$$= 0.5692 \text{ tCO}_2/\text{MWh}$$

The baseline emission factor is ex-ante parameter and will remain constant throughout the crediting period.

$EF_{grid, CM, y}$	=	Combined Margin Grid Emission Factor = 0.5692 tCO₂/MWh
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4.2 Project Emissions

The BESS will be charged with the electricity generated by the associated renewable power plant only, it is not charged using grid electricity or electricity from fossil fuel generator.

So as per the methodology, the project emission, $PE_y = 0$.

4.3 Leakage

No other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g., extraction, processing, transport etc.) are neglected

4.4 Net GHG Emission Reductions and Removals

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (t CO_{2e}/yr)

BE_y = Baseline emissions in year y (t CO₂/yr)

PE_y = Project emissions in year y (t CO_{2e}/yr)

Year	Estimated baseline emissions or removals (tCO _{2e})	Estimated project emissions or removals (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated net GHG emission reductions or removals (tCO _{2e})
Year 1	19020	0	0	19,020
Year 2	18917	0	0	18,917
Year 3	18814	0	0	18,814
Year 4	18713	0	0	18,713

Year 5	18611	0	0	18,611
Year 6	18510	0	0	18,510
Year 7	18410	0	0	18,410
Total	130995	0	0	130995
Total number of crediting years	7			
Annual Average over the crediting period	18713			

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,OM, y}
Data unit	tCO ₂ /MWh
Description	Operating margin CO ₂ emission factor of Thailand National Grid
Source of data	Report 'Thailand Grid Emission Factor for GHG Reduction Project/Activity' dated 28-09-2017 Published by Thailand Greenhouse Gas Management Organisation (Public Organisation)
Value applied	0. 5719
Justification of choice of data or description of measurement methods and procedures applied	Calculated in Report 'Thailand Grid Emission Factor for GHG Reduction Project/Activity' dated 28-09-2017 Published by Thailand Greenhouse Gas Management Organisation (Public Organisation) in line with "<i>Tool to calculate the emission factor for an electricity system</i>". The value used is calculated ex-ante as generation based weighted average of last three years of the operating margin provided in the Thailand Grid Emission Factor for GHG Reduction Project/Activity' Weighted average

	$= \sum_{i=1 \text{ to } n} (\text{Net generation in operating margin in year } i * \text{Simple operating margin in year } i) / \sum_{i=1 \text{ to } n} (\text{Net generation in operating margin of year } i)$
Purpose of Data	Baseline Emission calculation
Comments	The operating margin emission factor is a 3-year generation-weighted average (2014, 2015 & 2016). The operating Margin is calculated ex ante and fixed during the crediting period

Data / Parameter	$EF_{\text{grid,BM},y}$
Data unit	tCO ₂ /MWh
Description	Build margin CO ₂ emission factor of Thailand National grid
Source of data	Report 'Thailand Grid Emission Factor for GHG Reduction Project/Activity' dated 28/09/2017 Published by Thailand Greenhouse Gas Management Organisation (Public Organisation)
Value applied	0.5609
Justification of choice of data or description of measurement methods and procedures applied	Calculated in Report 'Thailand Grid Emission Factor for GHG Reduction Project/Activity' dated 28-09-2017 Published by Thailand Greenhouse Gas Management Organisation (Public Organisation) in line with "Tool to calculate the emission factor for an electricity system".
Purpose of Data	Baseline Emission calculation
Comments	The Build Margin would be calculated ex ante and fixed during the crediting period. For ex ante calculation the most recent data (2016) available has been used and the build margin is thus calculated.

Data / Parameter	$EF_{\text{grid,CM},y}$
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor of Thailand National grid

Source of data	Report 'Thailand Grid Emission Factor for GHG Reduction Project/Activity' dated 28-09-2017 Published by Thailand Greenhouse Gas Management Organization (Public Organization)
Value applied	0.5692
Justification of choice of data or description of measurement methods and procedures applied	The date has been considered in accordance to the Tool to calculate emission factor of an electricity system. The tool guides to take 75% weightage of $EF_{grid,OMsimple}$, & 25% weightage of $EF_{grid,BM,y}$.
Purpose of Data	Baseline Emission calculation
Comments	The combined margin would be calculated ex-ante and fixed for the entire crediting period and the combined margin thus calculated is 0.5692.

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/Year
Description	Quantity of net electricity generation supplied by the project (Solar) plant/unit to the grid in year y
Source of data	Monthly joint meter reading reports
Description of measurement methods and procedures to be applied	The difference of final value of export and import is used for monthly values of net electricity supplied to the grid by the project activity and same value will be considered for ER calculations.
Frequency of monitoring/recording	Continuous measurement & Monthly recording
Value applied	30,104
Monitoring equipment	Energy meters of accuracy class 0.2s
QA/QC procedures to be applied	The reading shall be crosschecked with the Invoice raised by the developer to consumer. The meter(s) shall be calibrated once in 5 years.

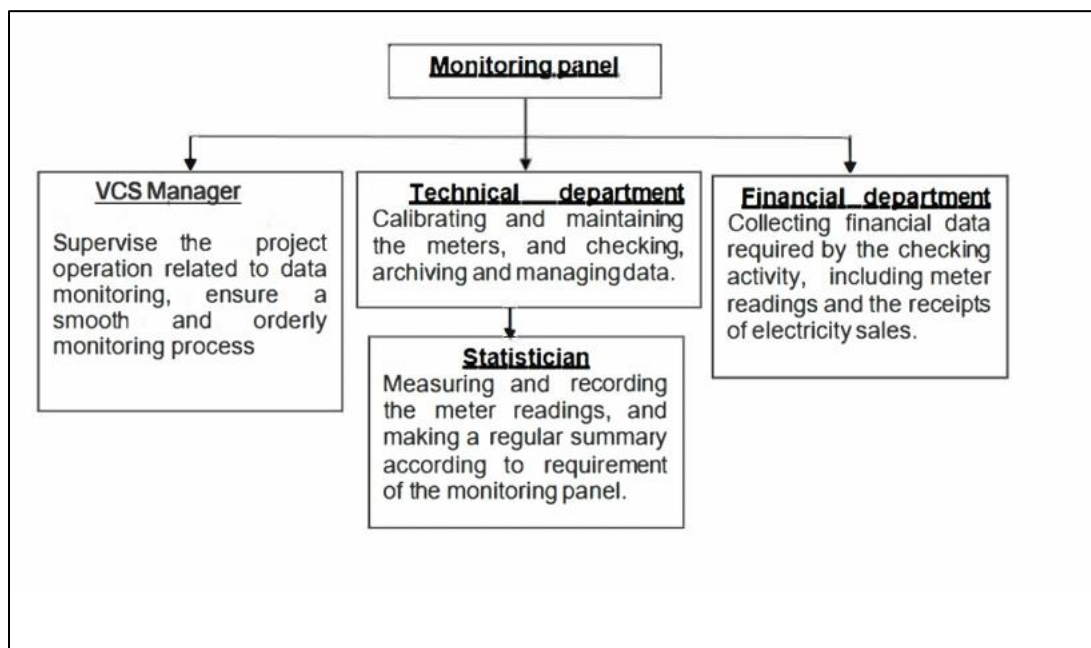
Purpose of data	Calculation of baseline emissions
Calculation method	Net electricity supplied to the grid by the project plant in a given month = Export(kWh)– Import (kWh)
Comments	Data will be archived in paper & electronic form for two years after the end of crediting period or of the last issuance of VERs for this project activity, whichever occurs later

5.3 Monitoring Plan

The purpose of the monitoring plan is to ensure that the monitoring and calculation of emission reductions of the project within the crediting period is complete, consistent, clear and accurate. The plan will be implemented by the project owner with the support of the grid corporation.

1. Monitoring organization

The monitoring process will be carried out and responsibility by the project owner. A monitoring panel will be established by the plant managers to be in charge of monitoring the data and information relating to the calculation of emission reductions with the cooperation of the Technical and Financial Department. A VCS manager will be assigned full charge the monitoring works. The operation and management structure is shown below:



Organization structure of the monitoring activity

2. Monitoring apparatus and installation

The meter(s) will be installed at the project site, to monitoring the input/output electricity at the grid side. The meter(s) will be installed in accordance with relevant national or international standard.

3. Data collection:

The specific steps for data collection and reporting are listed below:

- a. During the crediting period, both the grid company and the project owner will record the values displayed by the main meter.
- b. Simultaneously to step a), the project owner will both record the values displayed by the backup meters.
- c. The meters will be calibrated according to the relevant regulation and request of EGAT
- d. The main meter's readings will be cross-checked with record document confirmed by EGAT.
- e. The project owner and the grid company will record both output and input power readings from the main meter. These data will be used to calculate the amount of net electricity delivered to the grid.
- f. The project owner will be responsible of providing copies of record document confirmed by EGAT to the DOE for verification.
- g. If the reading of the main meter in a certain month is inaccurate and beyond the allowable error or the meter doesn't work normally, the grid-connected power generation shall be determined by following measures:
- h. Read the data of the backup meters.
- i. If the backup meter's data is not so accurate as to be accepted, or the practice is not standardized, the project owner and the grid corporation should jointly make a reasonable and conservative estimation method which can be supported by sufficient evidence and proved to be reasonable and conservative when verified by DOE.
- j. If the project owner and the grid corporation don't agree on an estimated method, arbitration will be conducted according the procedures set by the agreement to work out an estimation method.

4. Calibration

The calibration frequency of the monitoring meters will be once in 5 years. The accuracy of the monitoring meters will not less than 0.2s. Calibration of Meters should be implemented according to relevant standards and rules accepted by the grid company EGAT. After the examination, the meters should be sealed. The lift of the seals requires the presence of both the project owner and the grid company. One party must not lift the seals or fiddle with the meters without the presence of the other party.

All the meters installed shall be tested by a qualified metering verification institution commissioned jointly by the project owner and the grid company within 10 days after:

- a. Detection of a difference larger than the allowable error in the readings of both meters;
- b. The repair of all or part of meter caused by the failure of one or more parts to operate in accordance with the specifications.

5. Data management system

Physical document such as the plant electrical wiring diagram will be gathered with this monitoring plan in a single place. In order to facilitate auditors' access to project documents, the project materials and monitoring results will be indexed. All paper-based information will be stored by the technical department of the project owner and all the material will have a copy for backup. All data, including calibration records, will be kept until 2 years after the end of the total crediting period.

6. Monitoring Report

During the crediting period, at the end of each year, the monitoring officer shall produce a monitoring report covering the past monitoring period. The report shall be transmitted to the General Manager who will check the data and issue a final monitoring report in the name of the project participants. Once the final report is issued, it will be submitted to the DOE for verification.